

5. Differentiate

$$\frac{d}{dx} \left[\sqrt{x^3 + 5} \right]$$

Step 1: Rewrite the square root

$$\sqrt{x^3 + 5} = (x^3 + 5)^{1/2}$$

Step 2: Use the chain rule

If

$$\frac{d}{dx} [u^{1/2}] = \frac{1}{2} u^{-1/2} \frac{du}{dx},$$

where

$$u = x^3 + 5.$$

Then

$$\frac{du}{dx} = 3x^2.$$

Therefore,

$$\frac{d}{dx} \left[(x^3 + 5)^{1/2} \right] = \frac{1}{2} (x^3 + 5)^{-1/2} (3x^2).$$

Hence,

$$\boxed{\frac{3x^2}{2\sqrt{x^3 + 5}}}$$

6. Differentiate

$$\frac{d}{dx} \left[\frac{x}{x^2 + 3} \right]$$

Use the quotient rule:

$$\left(\frac{u}{v} \right)' = \frac{u'v - uv'}{v^2}.$$

Let

$$u = x, \quad v = x^2 + 3.$$

Then

$$u' = 1, \quad v' = 2x.$$

Substituting:

$$\begin{aligned} &= \frac{(1)(x^2 + 3) - x(2x)}{(x^2 + 3)^2} \\ &= \frac{x^2 + 3 - 2x^2}{(x^2 + 3)^2} \\ &= \boxed{\frac{3 - x^2}{(x^2 + 3)^2}}. \end{aligned}$$

7. Differentiate

$$\frac{d}{dx}[\sin(2\sqrt{x})]$$

Use the chain rule.

Let

$$u = 2\sqrt{x} = 2x^{1/2}.$$

Then

$$u' = 2 \left(\frac{1}{2} x^{-1/2} \right) = x^{-1/2} = \frac{1}{\sqrt{x}}.$$

Since

$$\frac{d}{dx}(\sin u) = \cos u \cdot u',$$

we get

$$= \cos(2\sqrt{x}) \cdot \frac{1}{\sqrt{x}}.$$

Therefore,

$$\boxed{\frac{\cos(2\sqrt{x})}{\sqrt{x}}}.$$

8. Differentiate

$$\frac{d}{dx}[\sin x - x \cos x]$$

Differentiate term by term.

First term

$$\frac{d}{dx}(\sin x) = \cos x.$$

Second term

Use the product rule:

$$\frac{d}{dx}(x \cos x) = (1)(\cos x) + x(-\sin x).$$

Thus

$$= \cos x - x \sin x.$$

Since the original expression is

$$\sin x - x \cos x,$$

its derivative is

$$\cos x - (\cos x - x \sin x).$$

Simplifying,

$$\boxed{x \sin x}.$$

9(a) Evaluate

$$\int x^8 dx$$

Use the power rule:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1.$$

Therefore,

$$\boxed{\frac{x^9}{9} + C}.$$

9(b)

$$\int x^{5/7} dx$$

Applying the power rule:

$$\begin{aligned} &= \frac{x^{12/7}}{12/7} + C \\ &= \boxed{\frac{7}{12}x^{12/7} + C}. \end{aligned}$$

9(c)

$$\int x^3 \sqrt{x} \, dx$$

First simplify:

$$x^3 \sqrt{x} = x^3 x^{1/2} = x^{7/2}.$$

Then

$$\int x^{7/2} \, dx = \frac{x^{9/2}}{9/2} + C.$$

Hence,

$$\boxed{\frac{2}{9}x^{9/2} + C}.$$

10(a)

$$\int \sqrt[3]{x^2} \, dx$$

Rewrite:

$$\sqrt[3]{x^2} = x^{2/3}.$$

Then

$$\int x^{2/3} \, dx = \frac{x^{5/3}}{5/3} + C.$$

Therefore,

$$\boxed{\frac{3}{5}x^{5/3} + C}.$$

10(b)

$$\int \frac{1}{x^6} dx$$

Rewrite:

$$\frac{1}{x^6} = x^{-6}.$$

Then

$$\int x^{-6} dx = \frac{x^{-5}}{-5} + C.$$

Thus,

$$\boxed{-\frac{1}{5x^5} + C}.$$

10(c)

$$\int x^{-7/8} dx$$

Using the power rule:

$$\begin{aligned} &= \frac{x^{1/8}}{1/8} + C \\ &= \boxed{8x^{1/8} + C}. \end{aligned}$$

Final Answers

Derivatives

5.

$$\boxed{\frac{3x^2}{2\sqrt{x^3 + 5}}}$$

6.

$$\boxed{\frac{3 - x^2}{(x^2 + 3)^2}}$$

7.

$$\boxed{\frac{\cos(2\sqrt{x})}{\sqrt{x}}}$$

8.

$$\boxed{x \sin x}$$

Integrals

9(a). $\boxed{\frac{x^9}{9} + C}$

$$9(b). \boxed{\frac{7}{12}x^{12/7} + C}$$

$$9(c). \boxed{\frac{2}{9}x^{9/2} + C}$$

$$10(a). \boxed{\frac{3}{5}x^{5/3} + C}$$

$$10(b). \boxed{-\frac{1}{5x^5} + C}$$

$$10(c). \boxed{8x^{1/8} + C}$$

Problems **11–14** using the **power rule of integration**:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1.$$

11.

$$\int \left(5x + \frac{2}{3x^5} \right) dx$$

Step 1: Rewrite using exponents

$$= \int \left(5x + \frac{2}{3}x^{-5} \right) dx$$

Integrate term by term:

$$= 5 \int x dx + \frac{2}{3} \int x^{-5} dx.$$

First term

$$5 \int x dx = 5 \left(\frac{x^2}{2} \right) = \frac{5x^2}{2}.$$

Second term

$$\frac{2}{3} \int x^{-5} dx = \frac{2}{3} \left(\frac{x^{-4}}{-4} \right) = -\frac{x^{-4}}{6}.$$

Rewrite without negative exponents:

$$-\frac{x^{-4}}{6} = -\frac{1}{6x^4}.$$

Therefore,

$$\int \left(5x + \frac{2}{3x^5} \right) dx = \frac{5x^2}{2} - \frac{1}{6x^4} + C.$$

12.

$$\int \left(x^{-1/2} - 3x^{7/5} + \frac{1}{5} \right) dx$$

Integrate each term separately.

First term

$$\int x^{-1/2} dx = \frac{x^{1/2}}{1/2} = 2x^{1/2}.$$

Second term

$$-3 \int x^{7/5} dx = -3 \left(\frac{x^{12/5}}{12/5} \right) = -\frac{15}{12} x^{12/5} = -\frac{5}{4} x^{12/5}.$$

Third term

$$\int \frac{1}{5} dx = \frac{x}{5}.$$

Combining,

$$2\sqrt{x} - \frac{5}{4}x^{12/5} + \frac{x}{5} + C.$$

13.

$$\int \left(x^{-3} - 3x^{1/4} + 8x^2 \right) dx$$

Integrate term by term.

First term

$$\int x^{-3} dx = \frac{x^{-2}}{-2} = -\frac{1}{2x^2}.$$

Second term

$$-3 \int x^{1/4} dx = -3 \left(\frac{x^{5/4}}{5/4} \right) = -\frac{12}{5} x^{5/4}.$$

Third term

$$8 \int x^2 dx = 8 \left(\frac{x^3}{3} \right) = \frac{8x^3}{3}.$$

Therefore,

$$\boxed{-\frac{1}{2x^2} - \frac{12}{5}x^{5/4} + \frac{8x^3}{3} + C}.$$

14.

$$\int \left[\frac{10}{y^{3/4}} - \sqrt[3]{y} + \frac{4}{\sqrt{y}} \right] dy$$

Rewrite using exponents:

$$= \int \left(10y^{-3/4} - y^{1/3} + 4y^{-1/2} \right) dy.$$

Integrate term by term.

First term

$$10 \int y^{-3/4} dy = 10 \left(\frac{y^{1/4}}{1/4} \right) = 40y^{1/4}.$$

Second term

$$- \int y^{1/3} dy = -\frac{y^{4/3}}{4/3} = -\frac{3}{4}y^{4/3}.$$

Third term

$$4 \int y^{-1/2} dy = 4 \left(\frac{y^{1/2}}{1/2} \right) = 8y^{1/2}.$$

Hence,

$$\boxed{40y^{1/4} - \frac{3}{4}y^{4/3} + 8\sqrt{y} + C}.$$

Final Answers

11.

$$\boxed{\frac{5x^2}{2} - \frac{1}{6x^4} + C}$$

12.

$$\boxed{2\sqrt{x} - \frac{5}{4}x^{12/5} + \frac{x}{5} + C}$$

13.

$$-\frac{1}{2x^2} - \frac{12}{5}x^{5/4} + \frac{8x^3}{3} + C$$

14.

$$40y^{1/4} - \frac{3}{4}y^{4/3} + 8\sqrt{y} + C$$

✨ A quick check: if you differentiate each answer, you'll recover the original integrand, confirming the results are correct.

Problems **15–24** step by step. ✨

15.

$$\int x(1 + x^3) dx$$

Step 1: Expand

$$x(1 + x^3) = x + x^4$$

Thus,

$$\int (x + x^4) dx = \int x dx + \int x^4 dx.$$

Step 2: Integrate

$$= \frac{x^2}{2} + \frac{x^5}{5} + C.$$

Answer

$$\frac{x^2}{2} + \frac{x^5}{5} + C$$

16.

$$\int (2 + y^2)^2 dy$$

Expand

$$(2 + y^2)^2 = 4 + 4y^2 + y^4.$$

Therefore,

$$\int (4 + 4y^2 + y^4) dy = 4y + \frac{4y^3}{3} + \frac{y^5}{5} + C.$$

Answer

$$4y + \frac{4y^3}{3} + \frac{y^5}{5} + C$$

17.

$$\int x^{1/3}(2-x)^2 dx$$

Expand

$$(2-x)^2 = 4 - 4x + x^2.$$

Multiply by $x^{1/3}$:

$$4x^{1/3} - 4x^{4/3} + x^{7/3}.$$

Integrate:

$$4 \left(\frac{x^{4/3}}{4/3} \right) - 4 \left(\frac{x^{7/3}}{7/3} \right) + \frac{x^{10/3}}{10/3} + C.$$

Simplify:

$$3x^{4/3} - \frac{12}{7}x^{7/3} + \frac{3}{10}x^{10/3} + C$$

18.

$$\int (1+x^2)(2-x) dx$$

Expand

$$(1+x^2)(2-x) = 2 - x + 2x^2 - x^3.$$

Integrate:

$$2x - \frac{x^2}{2} + \frac{2x^3}{3} - \frac{x^4}{4} + C.$$

Answer

$$2x - \frac{x^2}{2} + \frac{2x^3}{3} - \frac{x^4}{4} + C$$

19.

$$\int \frac{x^5 + 2x^2 - 1}{x^4} dx$$

Simplify

$$= x + \frac{2}{x^2} - \frac{1}{x^4} = x + 2x^{-2} - x^{-4}.$$

Integrate:

$$\frac{x^2}{2} + 2 \left(\frac{x^{-1}}{-1} \right) - \frac{x^{-3}}{-3} + C.$$

Thus,

$$\boxed{\frac{x^2}{2} - \frac{2}{x} + \frac{1}{3x^3} + C}$$

20.

$$\int \frac{1 - 2t^3}{t^3} dt$$

Simplify

$$= t^{-3} - 2.$$

Integrate:

$$\frac{t^{-2}}{-2} - 2t + C.$$

Therefore,

$$\boxed{-\frac{1}{2t^2} - 2t + C}$$

21.

$$\int \left(\frac{2}{x} + 3e^x \right) dx$$

Use:

$$\int \frac{1}{x} dx = \ln |x|$$

and

$$\int e^x dx = e^x.$$

Hence,

$$2 \ln |x| + 3e^x + C.$$

Answer

$$\boxed{2 \ln |x| + 3e^x + C}$$

22.

$$\int \left(\frac{1}{2t} - \sqrt{2} e^t \right) dt$$

Integrate term by term:

$$\frac{1}{2} \int \frac{1}{t} dt - \sqrt{2} \int e^t dt.$$

Therefore,

$$\boxed{\frac{1}{2} \ln |t| - \sqrt{2} e^t + C}$$

23.

$$\int (3 \sin x - 2 \sec^2 x) dx$$

Use:

$$\int \sin x dx = -\cos x, \quad \int \sec^2 x dx = \tan x.$$

Thus,

$$3(-\cos x) - 2(\tan x) + C.$$

Answer

$$\boxed{-3 \cos x - 2 \tan x + C}$$

24.

$$\int (\csc^2 t - \sec t \tan t) dt$$

Recall:

$$\int \csc^2 t dt = -\cot t,$$

because

$$\frac{d}{dt}(\cot t) = -\csc^2 t,$$

and

$$\int \sec t \tan t dt = \sec t.$$

Therefore,

$$-\cot t - \sec t + C.$$

Answer

$-\cot t - \sec t + C$

Final Answers

Problem	Answer
15	$\frac{x^2}{2} + \frac{x^5}{5} + C$
16	$4y + \frac{4y^3}{3} + \frac{y^5}{5} + C$
17	$3x^{4/3} - \frac{12}{7}x^{7/3} + \frac{3}{10}x^{10/3} + C$
18	$2x - \frac{x^2}{2} + \frac{2x^3}{3} - \frac{x^4}{4} + C$
19	$\frac{x^2}{2} - \frac{2}{x} + \frac{1}{3x^3} + C$
20	$-\frac{1}{2t^2} - 2t + C$
21	-
22	-
23	$-3\cos x - 2\tan x + C$
24	$-\cot t - \sec t + C$

✨ Notice that Problems **21–24** introduce the special antiderivatives:

$$\int \frac{1}{x} dx = \ln |x| + C, \quad \int e^x dx = e^x + C,$$

$$\int \sec x \tan x dx = \sec x + C.$$

These are worth memorizing because they appear very frequently in calculus.

25.

$$\int \sec x (\sec x + \tan x) dx$$

Step 1: Expand

Multiply:

$$\sec x (\sec x + \tan x) = \sec^2 x + \sec x \tan x.$$

Thus,

$$\int (\sec^2 x + \sec x \tan x) dx.$$

Step 2: Integrate each term

Using

$$\int \sec^2 x dx = \tan x,$$

and

$$\int \sec x \tan x dx = \sec x,$$

we obtain

$$\tan x + \sec x + C.$$

Therefore,

$\tan x + \sec x + C$

26.

$$\int \csc x (\sin x + \cot x) dx$$

Step 1: Expand

Since

$$\csc x = \frac{1}{\sin x},$$

we have

$$\csc x(\sin x + \cot x) = \csc x \sin x + \csc x \cot x.$$

Because

$$\csc x \sin x = 1,$$

this becomes

$$1 + \csc x \cot x.$$

Therefore,

$$\int (1 + \csc x \cot x) dx.$$

Step 2: Integrate

Recall

$$\int 1 dx = x,$$

and since

$$\frac{d}{dx}(\csc x) = -\csc x \cot x,$$

we have

$$\int \csc x \cot x dx = -\csc x.$$

Hence,

$$\boxed{x - \csc x + C}$$

27.

$$\int \frac{\sec \theta}{\cos \theta} d\theta$$

Step 1: Simplify

Since

$$\sec \theta = \frac{1}{\cos \theta},$$

we get

$$\frac{\sec \theta}{\cos \theta} = \frac{1}{\cos^2 \theta} = \sec^2 \theta.$$

Thus,

$$\int \sec^2 \theta \, d\theta.$$

Using

$$\int \sec^2 \theta \, d\theta = \tan \theta,$$

we obtain

$$\boxed{\tan \theta + C}$$

28.

$$\int \frac{dy}{\csc y}$$

Step 1: Simplify

Since

$$\csc y = \frac{1}{\sin y},$$

then

$$\frac{1}{\csc y} = \sin y.$$

Therefore,

$$\int \sin y \, dy.$$

Using

$$\int \sin y \, dy = -\cos y,$$

we get

$$\boxed{-\cos y + C}$$

Final Answers

Problem	Answer
23	$-3 \cos x - 2 \tan x + C$
24	$-\cot t - \sec t + C$
25	$\tan x + \sec x + C$
26	$x - \csc x + C$
27	$\tan \theta + C$
28	$-\cos y + C$

Useful Trigonometric Integrals to Remember

$$\int \sec^2 x \, dx = \tan x + C,$$

$\quad$

$$\int \csc x \cot x \, dx = -\csc x + C,$$

$$\int \sin x \, dx = -\cos x + C,$$

$\quad$

$$\int \cos x \, dx = \sin x + C.$$

These identities make problems like **23–28** very quick once memorized. 🌟

29.

$$\int \frac{\sin x}{\cos^2 x} \, dx$$

Step 1: Rewrite

Since

$$\frac{\sin x}{\cos^2 x} = \left(\frac{\sin x}{\cos x} \right) \left(\frac{1}{\cos x} \right),$$

we get

$$= \tan x \sec x.$$

Therefore,

$$\int \tan x \sec x \, dx.$$

Step 2: Integrate

Recall:

$$\frac{d}{dx}(\sec x) = \sec x \tan x.$$

Hence,

$$\boxed{\sec x + C}$$

30.

$$\int \left(\phi + \frac{2}{\sin^2 \phi} \right) d\phi$$

Rewrite:

$$\frac{2}{\sin^2 \phi} = 2 \csc^2 \phi.$$

So,

$$= \int \phi d\phi + 2 \int \csc^2 \phi d\phi.$$

Integrating,

$$\int \phi d\phi = \frac{\phi^2}{2},$$

and

$$\int \csc^2 \phi d\phi = -\cot \phi.$$

Thus,

$$\boxed{\frac{\phi^2}{2} - 2 \cot \phi + C}$$

31.

$$\int (1 + \sin^2 \theta \csc \theta) d\theta$$

Step 1: Simplify

Since

$$\csc \theta = \frac{1}{\sin \theta},$$

we have

$$\sin^2 \theta \csc \theta = \sin \theta.$$

Thus,

$$\int (1 + \sin \theta) d\theta.$$

Step 2: Integrate

$$\int 1 d\theta = \theta,$$

and

$$\int \sin \theta d\theta = -\cos \theta.$$

Therefore,

$$\boxed{\theta - \cos \theta + C}$$

32.

$$\int \frac{\sec x + \cos x}{2 \cos x} dx$$

Step 1: Split the fraction

$$= \frac{1}{2} \int \left(\frac{\sec x}{\cos x} + \frac{\cos x}{\cos x} \right) dx.$$

Since

$$\frac{\sec x}{\cos x} = \frac{1/\cos x}{\cos x} = \sec^2 x,$$

and

$$\frac{\cos x}{\cos x} = 1,$$

the integral becomes

$$\frac{1}{2} \int (\sec^2 x + 1) dx.$$

Step 2: Integrate

$$= \frac{1}{2} (\tan x + x) + C.$$

Hence,

$$\boxed{\frac{\tan x + x}{2} + C}$$

33.

$$\int \left[\frac{1}{2\sqrt{1-x^2}} - \frac{3}{1+x^2} \right] dx$$

Step 1: Recognize standard forms

Recall:

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C,$$

and

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C.$$

Step 2: Integrate

Therefore,

$$\frac{1}{2} \sin^{-1} x - 3 \tan^{-1} x + C.$$

Hence,

$$\frac{1}{2} \sin^{-1} x - 3 \tan^{-1} x + C$$

Final Answers

Problem	Answer
29	$\sec x + C$
30	$\frac{\phi^2}{2} - 2 \cot \phi + C$
31	$\theta - \cos \theta + C$
32	$\frac{\tan x + x}{2} + C$
33	$\frac{1}{2} \sin^{-1} x - 3 \tan^{-1} x + C$

Standard Integrals Used

$$\int \sec x \tan x \, dx = \sec x + C,$$

$$\int \csc^2 x \, dx = -\cot x + C,$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C,$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C.$$

These inverse trigonometric integrals are especially important and appear frequently in calculus problems.

34.

Evaluate

$$\int \left[\frac{4}{x\sqrt{x^2-1}} + \frac{1+x+x^3}{1+x^2} \right] dx$$

Step 1: Split the integral

$$= \int \frac{4}{x\sqrt{x^2-1}} dx + \int \frac{1+x+x^3}{1+x^2} dx.$$

Part A:

$$\int \frac{4}{x\sqrt{x^2-1}} dx$$

Recall the standard formula:

$$\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a} \sec^{-1} |x/a| + C.$$

Here,

$$a = 1,$$

so

$$\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} |x| + C.$$

Therefore,

$$\int \frac{4}{x\sqrt{x^2-1}} dx = 4 \sec^{-1} |x|.$$

Part B:

$$\int \frac{1 + x + x^3}{1 + x^2} dx$$

Use polynomial division.

Divide

$$x^3 + x + 1$$

by

$$x^2 + 1.$$

We get

$$\frac{x^3 + x + 1}{x^2 + 1} = x + \frac{1}{x^2 + 1}.$$

Thus,

$$\int \left(x + \frac{1}{x^2 + 1} \right) dx.$$

Integrating,

$$\frac{x^2}{2} + \tan^{-1} x.$$

Final Answer

Combining both parts,

$$4 \sec^{-1} |x| + \frac{x^2}{2} + \tan^{-1} x + C$$

35.

Evaluate

$$\int \frac{1}{1 + \sin x} dx$$

by multiplying the numerator and denominator by an appropriate expression.

Step 1: Rationalize the denominator

Multiply numerator and denominator by

$$1 - \sin x.$$

Then

$$\frac{1}{1 + \sin x} = \frac{1 - \sin x}{(1 + \sin x)(1 - \sin x)}.$$

Using

$$1 - \sin^2 x = \cos^2 x,$$

we get

$$= \frac{1 - \sin x}{\cos^2 x}.$$

Split:

$$= \frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x} = \sec^2 x - \sec x \tan x.$$

Step 2: Integrate

$$\int \sec^2 x \, dx = \tan x,$$

and

$$\int \sec x \tan x \, dx = \sec x.$$

Therefore,

$$\boxed{\tan x - \sec x + C}$$

Check

Differentiate:

$$\frac{d}{dx}(\tan x - \sec x) = \sec^2 x - \sec x \tan x = \frac{1}{1 + \sin x}.$$

Correct! 

36.

Use

$$\cos 2x = 2 \cos^2 x - 1$$

to evaluate

$$\int \frac{1}{1 + \cos 2x} \, dx.$$

Step 1: Simplify the denominator

Using

$$\cos 2x = 2 \cos^2 x - 1,$$

we have

$$1 + \cos 2x = 1 + (2 \cos^2 x - 1) = 2 \cos^2 x.$$

Thus,

$$\frac{1}{1 + \cos 2x} = \frac{1}{2 \cos^2 x} = \frac{1}{2} \sec^2 x.$$

Step 2: Integrate

Therefore,

$$\int \frac{1}{2} \sec^2 x \, dx = \frac{1}{2} \tan x + C.$$

Hence,

$$\boxed{\frac{1}{2} \tan x + C}$$

Final Answers

Problem	Answer
34	$\frac{1}{4} \sec^{-1}$
35	$\tan x - \sec x + C$
36	$\frac{1}{2} \tan x + C$

Key Ideas Learned ✨

- **Polynomial division** simplifies rational functions before integrating.
- Multiplying by a **conjugate expression** such as $1 - \sin x$ can transform difficult trigonometric integrals.
- **Double-angle identities** like

$$\cos 2x = 2 \cos^2 x - 1$$

are powerful tools for simplifying integrands.